

MADALIN-IULIAN MILITARU

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 [LinkedIn](#)

 [GitHub](#)

Education

Bachelor of Computer Science and Engineering, University POLITEHNICA of Bucharest Oct 2023 – Jun 2027

Relevant Coursework: Data Structures and Algorithms, Operating Systems, Object Oriented Programming, Algorithm Design

Projects

Web Client with REST API Communication

C++

- Developed a C++ client application to manage a movie and collection system by communicating with a server via **REST API**
- Implemented core functionalities—using **HTTP requests** and **JSON response processing**—to provide a command-line interface for both administrators and regular users

Museum Visitor Database Management

Java

- Designed and implemented a museum visitor database management system in **Java**
- Utilized **design patterns** (e.g., Factory, Singleton) to structure the application for efficient data handling and scalable object creation/interaction

Custom Memory Allocator

C

- Built an imitation of standard library memory management functions (**malloc**, **calloc**, **realloc**) in C
- Leveraged the **sbrk system call** to manage the program's data segment, demonstrating an in-depth understanding of low-level memory allocation techniques

Competitions

NASA Space Apps Challenge: Sentinel-1 SAR Data Processing

October 2025

- Executed processing of Sentinel-1 SAR imagery to monitor changes in wetland boundaries and soil moisture.

EESTEC Olympics 13: AI/ML SIEM for POS Fraud Alerting System

October 2025

- Developed an AI/ML model using **PyTorch** for real-time fraud detection in POS transactions.
- Implemented real-time data processing and parallel alert generation using **multi-threading**.
- Integrated the system with external services via **API** calls and **Server-Sent Events (SSE)** for continuous data ingestion and alerting.

Adobe: Innovate4AI Hackathon

November 2025

- Developed an **agentic AI application** that identifies the **highest-engagement** YouTube fashion content by calculating **popularity and trend scores** based on view metrics.
- Built an **automated fashion newsletter system** that aggregates top **fashion videos** into curated trend-based updates.
- Implemented a **fashion-focused chatbot** restricted to fashion-related questions, using **Google Gemini API**, **YouTube Data API**, a **SQLite** database, and **Docker** for deployment.

Skills

- Programming Languages:** C/C++, Java, Python
- Technologies/Tools:** Linux, Git/GitHub